K-Means Clustering

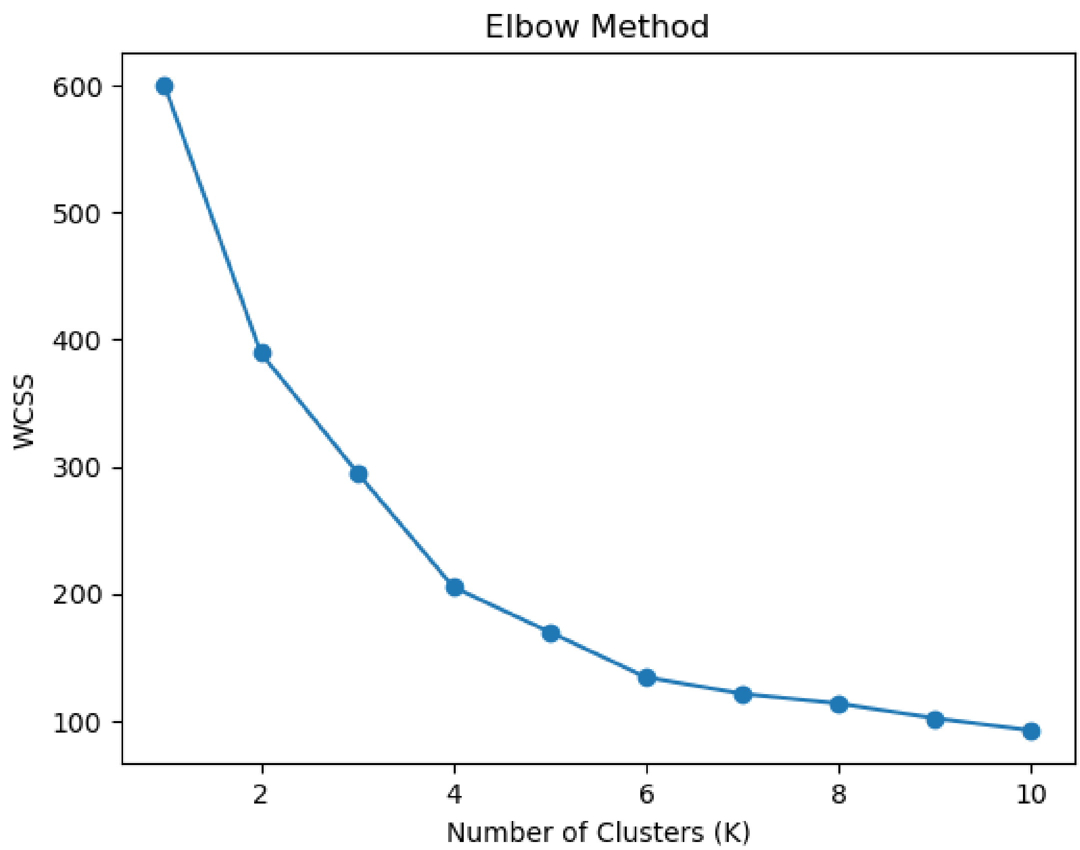
April 9, 2026

[2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.preprocessing import StandardScaler
```

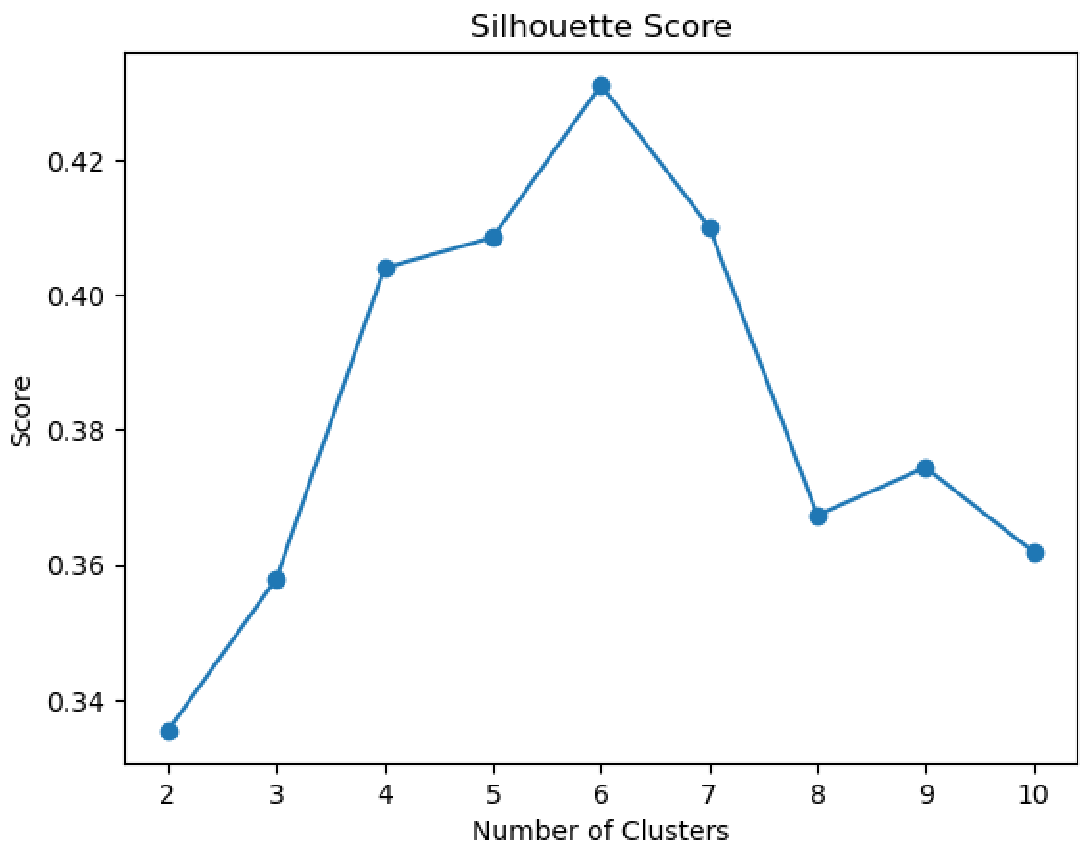
[6]:
```
data = pd.read_csv("Mall_Customers.csv")

# Select relevant features
X = data[['Annual Income (k$)', 'Spending Score (1-100)', 'Age']]
```

[8]:
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

[10]:
```
wcss = []  # Within Cluster Sum of Squares

for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

# Plot Elbow Curve
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.show()
```



[12]:
```
silhouette_scores = []

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    silhouette_scores.append(score)

# Plot
plt.plot(range(2, 11), silhouette_scores, marker='o')
plt.title("Silhouette Score")
plt.xlabel("Number of Clusters")
plt.ylabel("Score")
plt.show()
```



[14]:
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optimal_k = 5  # change based on above results

kmeans = KMeans(n_clusters=optimal_k, random_state=42)
data['Cluster'] = kmeans.fit_predict(X_scaled)
```

[16]:
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plt.figure(figsize=(8,6))

for i in range(optimal_k):
    plt.scatter(
        data[data['Cluster'] == i]['Annual Income (k$)'],
        data[data['Cluster'] == i]['Spending Score (1-100)'],
        label=f'Cluster {i}'
    )

plt.xlabel('Annual Income')
plt.ylabel('Spending Score')
plt.title('Customer Segments')
plt.legend()
plt.show()
```

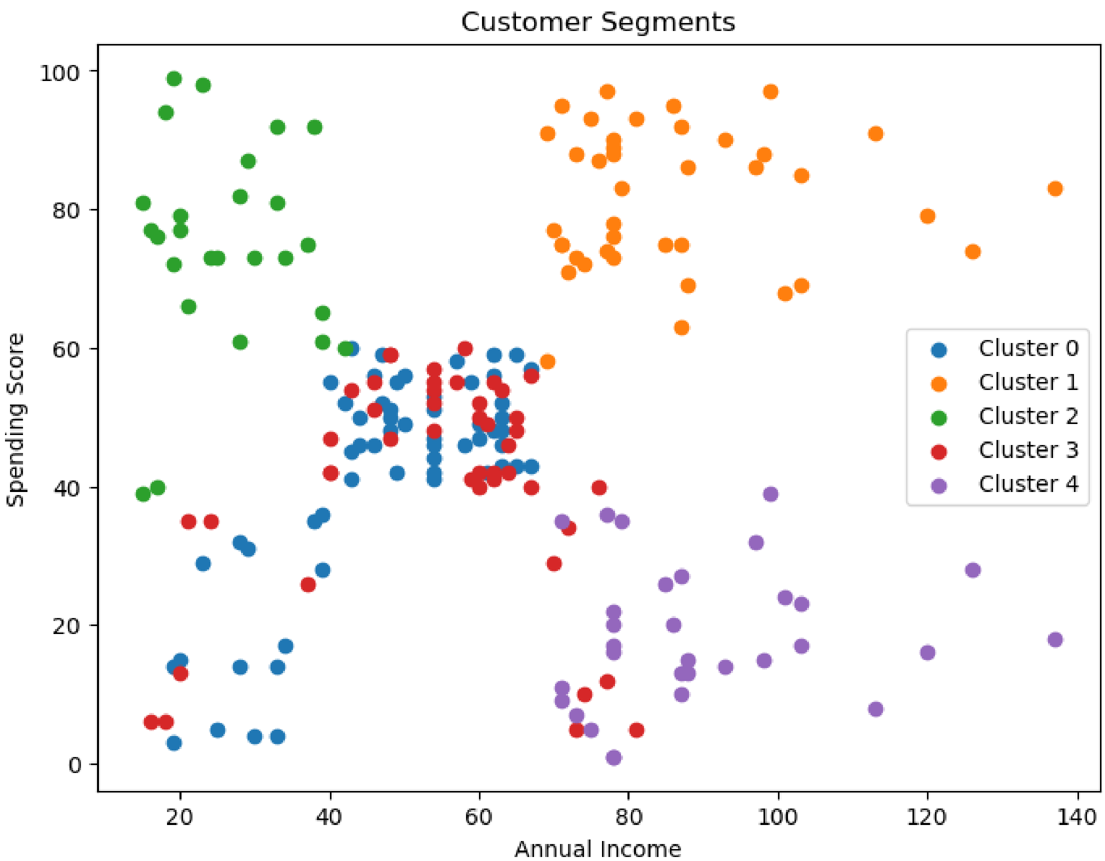

Customer Segments
Spending Score
Annual Income
Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
100
80
60
40
20
0
20
40
60
80
100
120
140


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[ ]:
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